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Editors

Nano Letters

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Dear Editors:

Please consider the enclosed Letter for publication in *Nano Letters*: “Mesoscale Arrest without Progressive Slowing in a Protein Polymer with Reversible β -Sheet-like Order.”

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Significance. Dynamic arrest in a soft material is usually described as a progressive slowing of motion. In the sequence-defined elastin-like polypeptide (VPAVG)₃₀ it proceeds differently. Small-angle X-ray photon correlation spectroscopy (SA-XPCS) at 30 °C resolves a two-step correlation decay whose fast-decay fraction falls from $f = 0.42\text{--}0.71$ to $f = 0.01\text{--}0.13$ over roughly 47 min, while the fast relaxation time shows no systematic drift and keeps a near-diffusive Q dependence. Material leaves the mobile population without slowing beforehand, and it does so at the largest length scales first: f rises monotonically with Q at every elapsed time, by nine standard deviations or more end to end across 76–167 nm. Arrest therefore proceeds by local conversion rather than bulk stiffening, which implies that a partially arrested protein network still contains fully mobile regions. Temperature-dependent SAXS/WAXS on the same construct resolves a reversible transition to a hierarchical assembly carrying β -sheet-like peaks at 1.38 and $0.72\text{ \AA}^{-1}$ that vanish on cooling, with seven consecutive 6–34 °C cycles recovering the low-temperature profile to within 9%.

Novelty. Prior work on gel-forming ELPs established arrested phase separation and network morphology from time-averaged scattering and rheology. The dynamical signature reported here, a diffusive fast mode whose time scale plateaus while its amplitude declines, has also been observed in a thermoreversible nanoparticle gel [D. Bahadur *et al.*, *J. Chem. Phys.* **151**, 104902 (2019)], and the manuscript cites that agreement rather than claiming the phenomenology as new. What is new is the length-scale decomposition. Fitting the fast fraction separately in each Q bin shows that arrest is ordered by length scale, reaching 167 nm before 76 nm at every elapsed time, and follows the transfer to near-completion at $f = 0.01$. That decomposition requires microsecond framing: with the fast relaxation time between 0.1 and 1.6 ms, a kilohertz acquisition returns only the slow component. We are equally explicit about the limits. The structural and dynamical observations come from separate experiments, so no causal link is claimed. The β -sheet assignment rests on WAXS peak positions rather than spectroscopy. The 1.6 s maximum delay bounds a fraction arrested on that scale, not an infinite-time nonergodicity factor. The isothermal series comes from one loaded sample. All four are stated in the manuscript, and the ordered contacts are called plausible rather than established junctions.

Impact and urgency. The work spans sequence-defined nanomaterials, protein self-assembly, nonequilibrium soft matter, and coherent X-ray characterization, and gives a nanoscale structure–dynamics framework for responsive protein materials whose β -sheet-like order switches thermally without permanent aggregation. Because the behavior is sequence-encoded, the strategy transfers to other programmable protein polymers. The result rests on a capability that is not widely available. The XSPA-500k at 8-ID-I frames at 50 kHz, among the fastest in routine operation at a storage ring. The fast relaxation reported here, 0.1 to 1.6 ms, would fall at or below the shortest delay of a conventional kilohertz acquisition, and the amplitude transfer that is the paper’s central result would not be visible at all. High-speed detectors and upgraded storage rings have only recently made this practical on weakly scattering biological samples, so a concise, carefully bounded demonstration is timely. The reduced data and all analysis and figure-generation code are already public on GitHub (qzhang234/PubCode-2021-IDPP).

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The manuscript is original, is not under consideration elsewhere, and has been approved by all authors. The authors declare no competing financial interest; AI-tool use is disclosed in the Acknowledgments. Thank you for your consideration.

Sincerely,

Qingteng Zhang, Ph.D., on behalf of all authors

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Suggested Reviewers

The following researchers have relevant expertise in elastin-like polypeptides, β -sheet-forming protein and peptide assemblies, small-angle X-ray scattering, and X-ray photon correlation spectroscopy.

Bradley D. Olsen

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Expertise: elastin-like polypeptide gels, associative protein polymers, arrested phase separation, and neutron spin-echo measurement of relaxation in protein hydrogels.

Darrin J. Pochan

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Expertise: peptide and protein self-assembly, β -sheet hydrogel networks, and small-angle scattering and rheology of soft biomaterials.

Andrei Fluerasu

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Expertise: X-ray photon correlation spectroscopy and the microscopic dynamics of gels and arrested soft matter.

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Expertise: small-angle X-ray scattering and X-ray photon correlation spectroscopy of colloids and soft matter.